

Goldbag Project Specification Document

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1. Executive Summary

1.1 Project Overview

[Goldbag](#) is a mobile-first fintech application that enables users to buy, sell, and invest in tokenised gold. The platform offers a seamless digital experience for gold investment, eliminating the complexities of physical storage by combining traditional precious metal investment with modern digital asset management features.

1.2 Project Objectives

- Create an accessible platform for gold investment with low entry barriers
 - Provide real-time gold trading capabilities with instant settlement
 - Enable social features to drive user engagement and viral growth
 - Offer yield-generating opportunities through gold lock-up mechanisms
 - Build a secure, compliant, and user-friendly investment platform
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2. Product Vision & Target Audience

2.1 Target Users

- **Primary:** Millennial and Gen-Z investors seeking alternative investments
- **Secondary:** Traditional investors looking to diversify portfolios digitally
- **Tertiary:** Crypto-savvy users interested in asset-backed tokens

2.2 Value Proposition

- Fractional gold ownership starting from minimal amounts
- No storage or insurance fees
- Instant liquidity with 24/7 trading
- Social investing features and referral rewards
- Earn interest on gold holdings

3. Core Features & Requirements

3.1 User Authentication & Onboarding

3.1.1 Registration

- Email and phone number verification
- Password creation with strength requirements
- KYC/AML compliance workflow
 - Identity verification (government-issued ID)
 - Address verification
 - Facial recognition/liveness check
- Terms of service and risk disclosure acceptance

3.1.2 Security

- Two-factor authentication (2FA)
- Biometric login (fingerprint/Face ID)

3.2 Wallet Management

3.2.1 GBP Wallet

- Display current GBP balance
- Transaction history with filtering and search
- Pending transactions indicator

3.2.2 Gold Holdings

- Display total gold owned (in grams/ounces/troy ounces)
- Current market value in GBP
- Average purchase price
- Total quantity breakdown

3.2.3 Funding Options

Requirements:

- Credit/debit card integration
- Apple Pay integration
- Minimum and maximum deposit limits
- Instant deposit confirmation for card payments
- Processing time indicators for each method

3.2.4 Withdrawal Options

Requirements:

- Withdraw to bank account
- Withdraw to payment card
- Withdrawal limits (daily/weekly/monthly)
- Processing time: 1-3 business days
- Withdrawal fee structure clearly displayed
- Bank account verification process

3.3 Gold Trading

3.3.1 Buy Gold

Requirements:

- Real-time gold price display (updates every minute)
- Input methods: GBP amount or gold quantity
- Live calculation of the counter value
- Order preview before confirmation
- Instant settlement upon confirmation
- Transaction confirmation screen
- Email and push notification of purchase

3.3.2 Sell Gold

Requirements:

- Current holdings display
- Input methods: GBP value or gold quantity to sell
- Display of applicable fees/spread
- Sell preview before confirmation
- Instant settlement upon confirmation
- Proceeds added to GBP wallet
- Transaction confirmation screen
- Email/push notification of sale

3.3.3 Price Chart

Requirements:

- Interactive candlestick/line chart
- Time intervals: 24H, 1W, 1M, 1Y
- Current price highlight
- Price change percentage display

3.4 Portfolio & PNL Tracking

3.4.1 Portfolio Overview

Requirements:

- Total portfolio value (GBP)
- Gold holdings value
- Available GBP balance
- Total profit/loss (GBP and percentage)
- Today's change (GBP and percentage)
- Portfolio allocation visualisation

3.4.2 PNL Details

Requirements:

- Unrealised profit/loss on current holdings
- Realised profit/loss from closed positions
- Average buy price vs. current price
- Return on investment (ROI) percentage
- Historical PNL chart
- Breakeven price indicator

3.4.3 Transaction History

Requirements:

- Complete transaction log (buys, sells, transfers, deposits, withdrawals)
- Filterable by transaction type and date range
- Searchable transaction history
- Individual transaction details view

3.5 Social Features

3.5.1 PNL Sharing

Requirements:

- Generate shareable PNL card/image
- Customizable share card design
- Share to: Twitter/X, Instagram, Facebook, LinkedIn, WhatsApp
- Copy link functionality
- Save image to device

3.5.2 Referral Program

Requirements:

- Unique referral code for each user
- Track referral status (pending, completed)
- Reward structure:
 - Referrer reward: X grams of gold when the referee completes the first purchase
 - Referee reward: Y grams of gold on first purchase
- Referral dashboard showing:
 - Total referrals
 - Pending referrals
 - Total rewards earned
 - Referral leaderboard (optional)
- Share referral code via multiple channels

3.6 Gold Transfer (P2P)**3.6.1 Send Gold****Requirements:**

- Search users by username
- Input gold amount to send
- GBP equivalent display
- Add optional message/note
- Confirmation screen with recipient details
- Transaction receipt
- Push notification to the recipient

3.6.2 Receive Gold**Requirements:**

- Notification of incoming transfer
- View sender details and message
- Automatic addition to gold holdings
- The transaction appears in history

3.7 Gold Savings (Lock & Earn) [Not required]**3.7.1 Savings Plans****Requirements:**

- Multiple lock-up periods available:

- 1M, 3M, 6M, 1Y
- Interest rates displayed for each period (APY)
- Minimum and maximum lock amounts
- Lock start and end date display
- Interest calculation methodology disclosure

3.7.2 Lock Gold

Requirements:

- Select gold amount to lock
- Choose the lock period
- Calculate projected interest earnings
- Terms and conditions acknowledgement
- Confirmation of lock-up
- Email confirmation with lock details

3.7.3 Active Savings

Requirements:

- Dashboard showing all active locks
- For each lock display:
 - Locked amount
 - Lock period
 - Start date and maturity date
 - Interest rate (APY)
 - Accrued interest to date
 - Projected total return
 - Progress indicator
- Early withdrawal option (with penalty disclosure)

3.7.4 Maturity & Interest Payout

Requirements:

- Automatic unlock on maturity date
- Interest paid in gold (added to holdings)
- Notification 7 days before maturity
- Notification on maturity day
- Option to automatically re-lock on maturity
- Maturity history log

3.8 Notifications

3.8.1 Push Notifications

- Transaction confirmations
- Price alerts (optional, user-configured)
- Savings maturity reminders
- Referral reward notifications
- Security alerts (new login, password change)
- Promotional messages (optional, opt-in)

3.8.2 Email Notifications

- Transaction confirmations
- Monthly portfolio statements
- Security alerts
- Savings maturity notifications
- Regulatory/compliance communications

3.8.3 In-App Notifications

- Centralized notification center
 - Read/unread status
 - Categorization by type
 - Archive functionality
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4. Technical Requirements

4.1 Platform

- iOS & Android

4.2 Technology Stack Recommendations

- **Frontend:** Flutter/React Native
- **Backend:** Node.js or Go
- **Database:** PostgreSQL (primary), Redis (caching)
- **Cloud Infrastructure:** AWS, Google Cloud, or Azure
- **Payment Processing:** Stripe, Plaid, or similar PCI-compliant processor
- **Authentication:** OAuth 2.0, JWT tokens
- **KYC/AML:** Integration with providers like Jumio, Onfido, or Persona

4.3 Performance Requirements

- App launch time: < 2 seconds
- API response time: < 500ms for 95th percentile

- Real-time price updates: 1-minute intervals
- Support for 10,000+ concurrent users
- 99.9% uptime SLA

4.4 Security Requirements

- End-to-end encryption for sensitive data
- PCI DSS compliance for payment processing
- SOC 2 Type II certification
- Regular security audits and penetration testing
- DDoS protection
- Rate limiting on API endpoints
- Secure key management (HSM or cloud KMS)
- Data encryption at rest and in transit

4.5 Compliance & Regulatory

- KYC/AML compliance (jurisdiction-specific)
- Anti-fraud detection system
- Transaction monitoring and suspicious activity reporting
- Data privacy compliance (GDPR, CCPA)
- Financial services licensing (as required)
- Audit trail for all transactions
- Terms of service and privacy policy
- Risk disclosures

4.6 Data Management

- User data retention policies
- Secure data backup and disaster recovery
- Data export functionality for users
- Right to deletion (GDPR compliance)

5. Integration Requirements

5.1 Third-Party Services

5.1.1 Payment Processors

- Credit/debit card processing
- Bank transfer processing
- Apple Pay integration

- Google Pay integration (Android)

5.1.2 Gold Price Data

- Real-time gold price feed (e.g., LBMA, COMEX)
- Fallback price sources for redundancy
- Historical price data for charting

5.1.3 Banking/Financial

- Bank account verification service
- Bank transfer processing

5.1.4 Identity Verification

- KYC provider integration
- Document verification
- Facial recognition
- Address verification

5.1.5 Communication

- Email service provider (SendGrid, AWS SES)
- Push notification service (FCM)

5.1.6 Analytics & Monitoring

- Application performance monitoring (Datadog, New Relic)
 - User analytics (Firebase Analytics)
 - Error tracking (Firebase Crashlytics)
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6. User Interface & Experience

6.1 Design Principles

- Clean, modern, and intuitive interface similar to Robin Hood & Partyvest
- Consistent design language across all screens
- Accessible colour contrast and typography
- Support for light and dark modes
- Accessibility compliance (WCAG 2.1 AA)

6.2 Key Screens

1. **Onboarding Flow** - Registration, KYC, tutorial
2. **Home/Dashboard** - Portfolio overview, quick actions
3. **Buy/Sell** - Trading interface with live prices
4. **Wallet** - GBP and gold balances, funding options
5. **Portfolio** - Detailed holdings, PNL, charts
6. **Savings** - Lock gold interface, active savings dashboard
7. **Transfer** - Send gold to other users
8. **Profile** - Settings, security, documents, support
9. **Referrals** - Referral dashboard and sharing
10. **Notifications** - Notification center

6.3 Navigation

- Bottom navigation bar (primary sections)
 - Hamburger menu or profile icon for secondary features
 - Deep linking support for notifications and shared content
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7. Business Logic

7.1 Gold Backing

To be defined:

- Physical gold reserves vs. gold futures vs. ETF backing
- Custody arrangements
- Proof of reserves mechanism
- Independent auditing

7.2 Pricing & Fees

To be defined:

- Buy/sell spread percentage
- Deposit fees (card, bank transfer)
- Withdrawal fees
- Transfer fees (P2P)
- Savings plan fees (if any)
- Revenue model and profitability targets

7.3 Interest Calculation

To be defined:

- Source of interest payments
- Interest rate determination methodology
- Compounding frequency
- Sustainability model

7.4 Limits & Controls

To be defined:

- Minimum buy/sell amounts
 - Maximum transaction sizes
 - Daily/weekly/monthly limits
 - Withdrawal limits
 - KYC tier-based limits
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8. Testing Requirements

8.1 Testing Types

- Unit testing (80%+ code coverage)
- Integration testing
- End-to-end testing
- Security testing and penetration testing
- Load testing and stress testing
- User acceptance testing (UAT)
- Beta testing with a limited user group

8.2 Test Environments

- Development
 - Staging (mirrors production)
 - Production
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9. Launch & Deployment

9.1 MVP Features (Phase 1)

- User registration and KYC
- GBP wallet with Apple Pay and Card funding

- Buy and sell gold
- Basic portfolio view with PNL
- Price chart
- Secure storage and basic security features

9.2 Phase 2 Features

- Gold savings (lock & earn)
- P2P gold transfers
- Enhanced portfolio analytics
- Bank withdrawal support

9.3 Phase 3 Features

- Referral program
 - PNL social sharing
 - Advanced charting and analytics
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10. Support & Maintenance

10.1 Customer Support

- In-app chat support
- Email support
- Help centre/FAQ
- Support ticket system
- Target response time: < 24 hours

10.2 Ongoing Maintenance

- Regular security updates
 - Bug fixes and performance optimisation
 - Feature enhancements based on user feedback
 - Regulatory compliance updates
 - Server and infrastructure maintenance
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11. Success Metrics & KPIs

11.1 User Metrics

- Monthly Active Users (MAU)
- Daily Active Users (DAU)
- User retention rate (D1, D7, D30)
- Customer acquisition cost (CAC)
- Lifetime value (LTV)

11.2 Business Metrics

- Total Assets Under Management (AUM)
- Transaction volume (GBP)
- Revenue (fees, spreads)
- Average transaction size
- Conversion rate (registration to first purchase)

11.3 Engagement Metrics

- Average session duration
- Transactions per user per month
- Referral conversion rate
- Social sharing rate
- Savings plan adoption rate

11.4 Technical Metrics

- App crash rate (< 1%)
 - API error rate (< 0.5%)
 - Average page load time
 - Uptime percentage
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12. Risk Assessment

12.1 Technical Risks

- Security breaches or hacks
- System downtime or outages
- Data loss
- Scalability challenges

Mitigation:

- Robust security architecture
- Regular security audits

- Redundant systems and backups
- Load testing and capacity planning

12.2 Regulatory Risks

- Licensing requirements
- Compliance violations
- Changes in regulations

Mitigation:

- Legal counsel engagement
- Compliance team
- Regular regulatory reviews
- Flexible architecture for regulatory changes

12.3 Market Risks

- Gold price volatility
- Liquidity issues
- Competition

Mitigation:

- Clear risk disclosures
- Diversified liquidity sources
- Unique value proposition and features

12.4 Operational Risks

- Fraud and money laundering
- Customer disputes
- Third-party service failures

Mitigation:

- Advanced fraud detection
- Clear terms and dispute resolution process
- Multiple service provider options

13. Timeline & Milestones

13.1 Estimated Project Timeline

Note: Specific dates to be determined based on resource availability

- **Weeks 1-2:** Requirements finalisation and design kickoff
- **Weeks 3-6:** UI/UX design and prototype
- **Weeks 7-18:** MVP development (Phase 1)
- **Weeks 19-20:** Internal testing and QA
- **Weeks 21-22:** Beta testing with select users
- **Week 23:** App store submission
- **Week 24-25:** Launch and monitoring
- **Weeks 26-38:** Phase 2 development
- **Weeks 39+:** Phase 3 and ongoing enhancements

13.2 Key Milestones

- ☐ Requirements approval
 - ☐ Design approval
 - ☐ MVP development complete
 - ☐ Security audit passed
 - ☐ Beta launch
 - ☐ Official launch
 - ☐ 1,000 users milestone
 - ☐ 10,000 users milestone
 - ☐ Break-even point
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14. Budget Considerations

Note: Detailed budget to be provided separately

14.1 Development Costs

- Design and UX
- Frontend and backend development
- QA and testing
- Project management

14.2 Infrastructure Costs

- Cloud hosting
- Database and storage
- CDN and bandwidth
- Development and staging environments

14.3 Third-Party Services

- Payment processing fees
- KYC/AML service fees
- SMS and email services
- Analytics and monitoring tools
- Gold price data feeds

14.4 Compliance & Legal

- Legal counsel
- Licensing fees
- Compliance services
- Insurance

14.5 Marketing & Launch

- App store optimisation
 - User acquisition campaigns
 - Referral program incentives
 - PR and communications
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15. Appendices

Appendix A: Glossary

- **Tokenised Gold:** Digital representation of physical gold ownership
- **KYC:** Know Your Customer - identity verification process
- **AML:** Anti-Money Laundering regulations
- **PNL:** Profit and Loss
- **APY:** Annual Percentage Yield
- **2FA:** Two-Factor Authentication

This document is confidential and proprietary to Goldbag. Distribution outside the project team requires approval.